

Calculated Fields, Parameters & LOD Expressions

QUESTION 1

What is a Calculated Field and why is it important in business dashboards?

ANSWER

A calculated field is a field (or column) that you create using a formula based on existing data in your dataset. It is not stored in the original data source—it is computed dynamically inside tools like Tableau or Microsoft Power BI.

It allows you to create new values from existing fields using formulas.

Example:

Profit = Sales – Cost

Profit Margin = Profit / Sales

These fields don't exist in the raw data—you create them for analysis.

Why is it Important in Business Dashboards?

Calculated fields are essential because dashboards are meant to show insights, not just raw data.

1. Converts Raw Data into Insights

Raw data: Sales, Cost

Insight: Profit, Growth Rate

2 . Helps Track KPIs (Key Performance Indicators)

Calculated fields are used to build KPIs like:

		Profit Margin							
		Year-to-Date (YTD) Sales							
		Average Order Value							
		3. Enables Better Decision-Making							
		Identify high-performing products							
		Detect loss-making regions							
		Compare performance across time							
		4 . Real-Time and Dynamic Analysis							
		Automatically updates when data changes							
		No need to manually recalculate values							
		5 . Improves Dashboard Visuals							
		Allows creation of meaningful charts (profit trends, growth analysis)							
		Makes dashboards more interactive and informative							

QUESTION 2		Why can't parameters change visuals directly in Tableau?				
ANSWER		parameters are NOT connected to data automatically				
		They don't filter data				
		They don't change charts by themselves				
		They only store a value				
		How Parameters Actually Work				
		To make a parameter affect a visual, you must connect it using a calculated field.				
		Suppose you create a parameter:				
		"Select Measure" (Sales / Profit)				
		Flexibility				
		You can use one parameter for multiple calculations				
		Not limited to simple filtering				
		Advanced Logic Support				
		Enables dynamic switching (measures, dimensions, calculations)				
		Useful for complex dashboards				
		Reusability				
		One parameter can control:				
		Charts				
		Filters				

		Titles					
		Calculations					
		Pressing the button alone does nothing					
		It needs to be connected to a device (calculated field) to work					
		Parameters = user input only					
		They don't directly interact with visuals					
		They must be linked using calculated fields or filters					

		Level of Detail							
		FIXED → You define exact level							
		INCLUDE → Adds more granularity to the existing level							
		Filter Behavior							
		FIXED → Ignores normal filters (unless added to context)							
		INCLUDE → Responds to filters							

QUESTION 5

Task 1: Delivery Performance Classification (Calculated Field)

Business Need

Operations wants to track delivery efficiency.

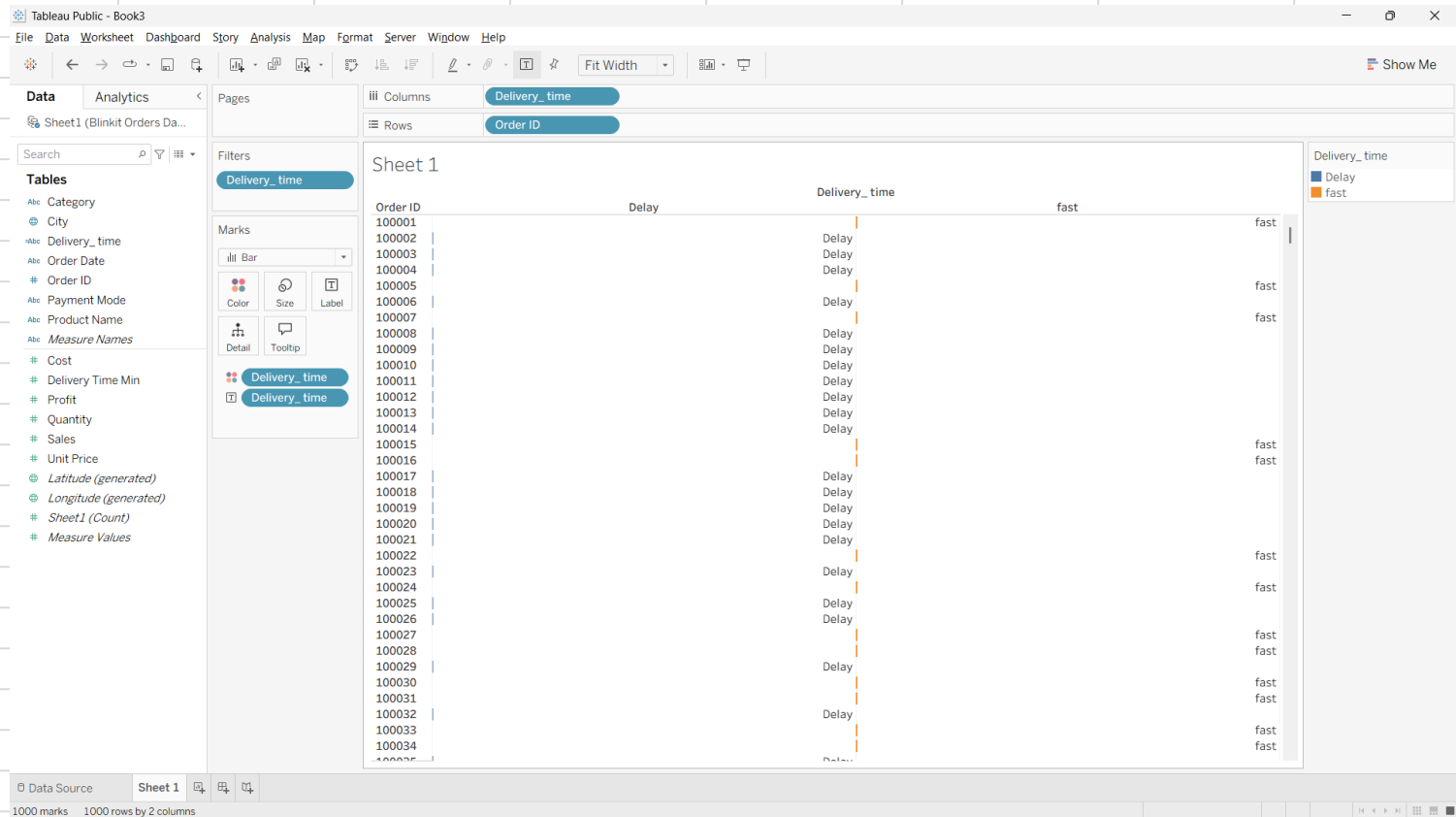
Task

Classify orders as Fast or Delayed

Condition $\text{Delivery} \leq 30 \text{ min} \rightarrow \text{Fast}$

Else $\rightarrow \text{Delayed}$

ANSWER



QUESTION 6

Task 2: Loss-Making Orders Identification

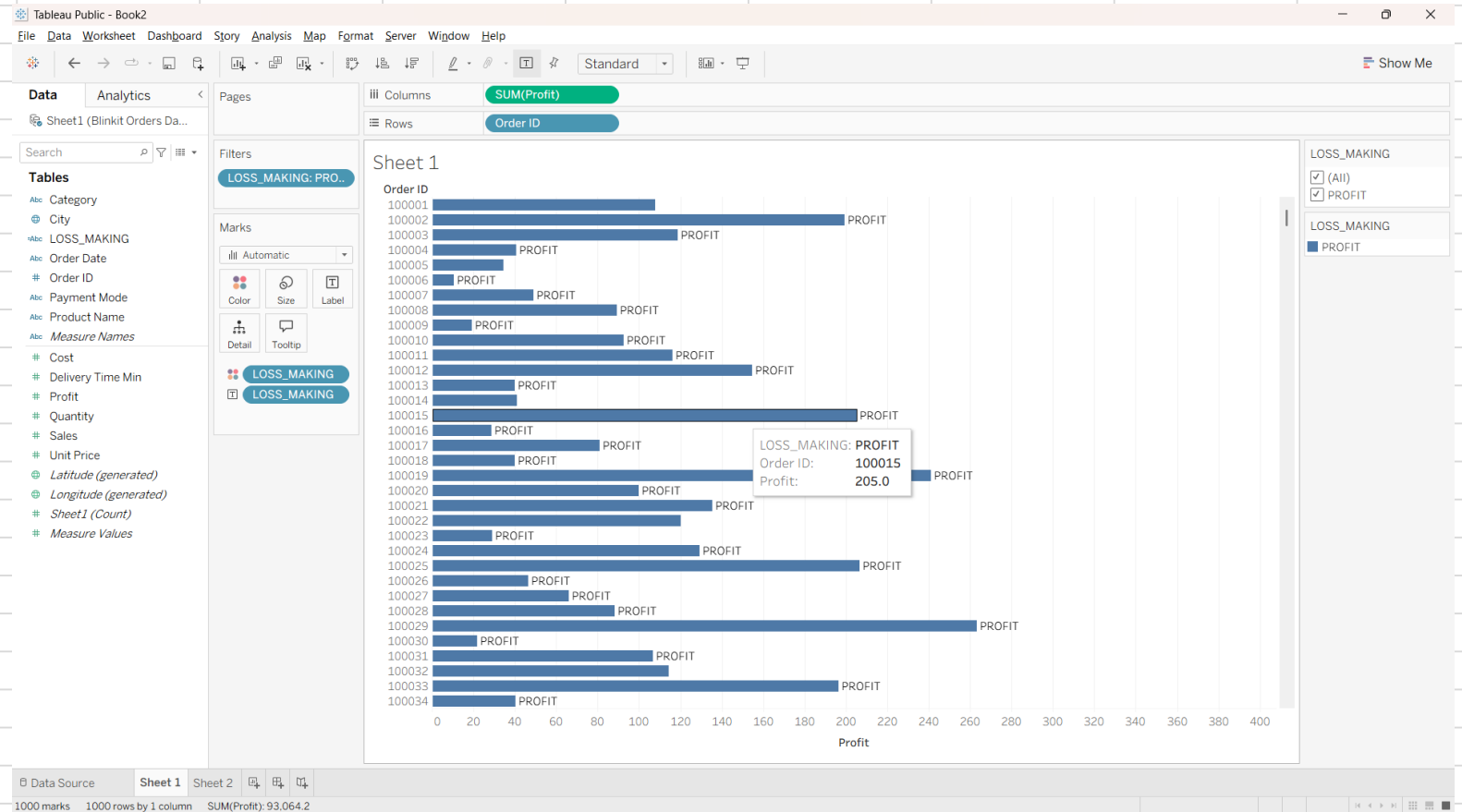
Business Need

High sales do not always mean high profit.

Task

Identify orders resulting in loss

ANSWER



QUESTION

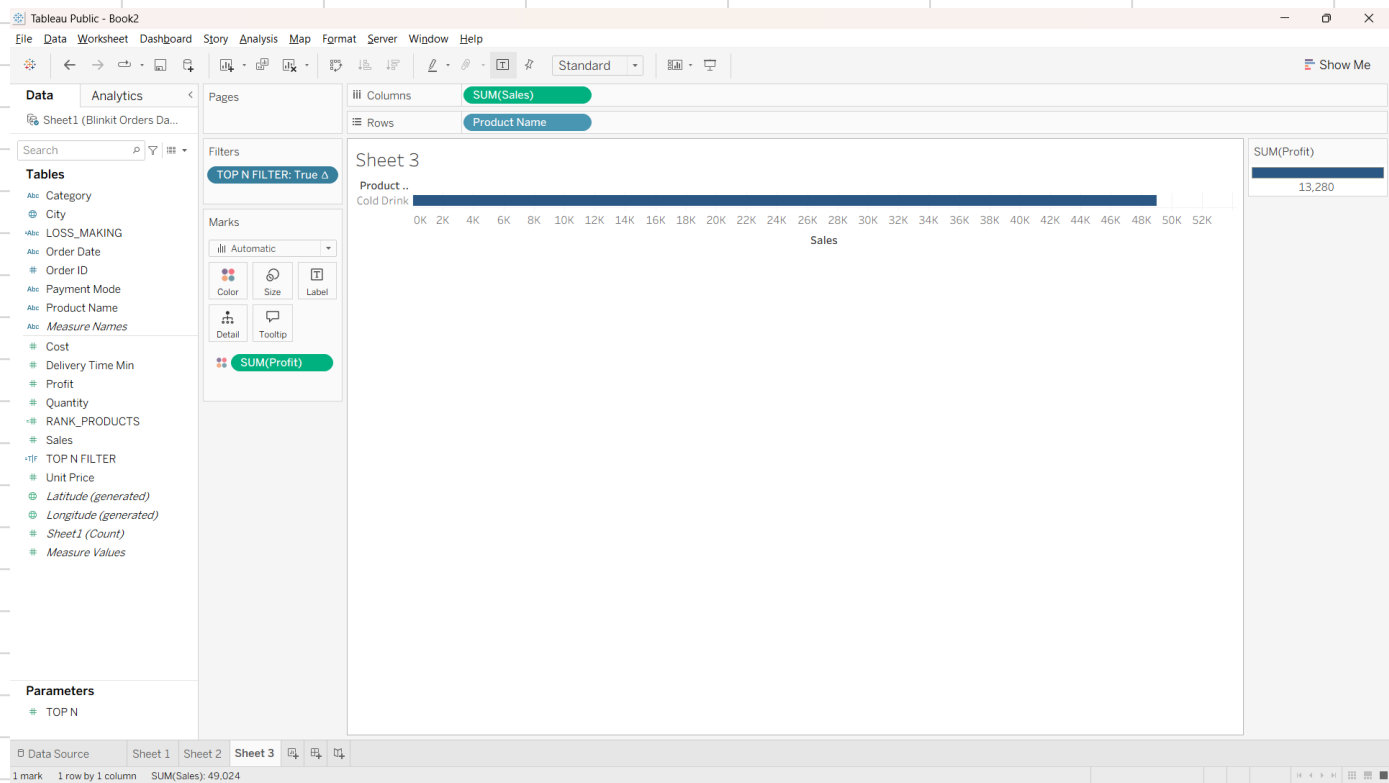
Task 3: Top N Products Analysis (Parameter Business Need)

The product team wants to focus on top-performing products

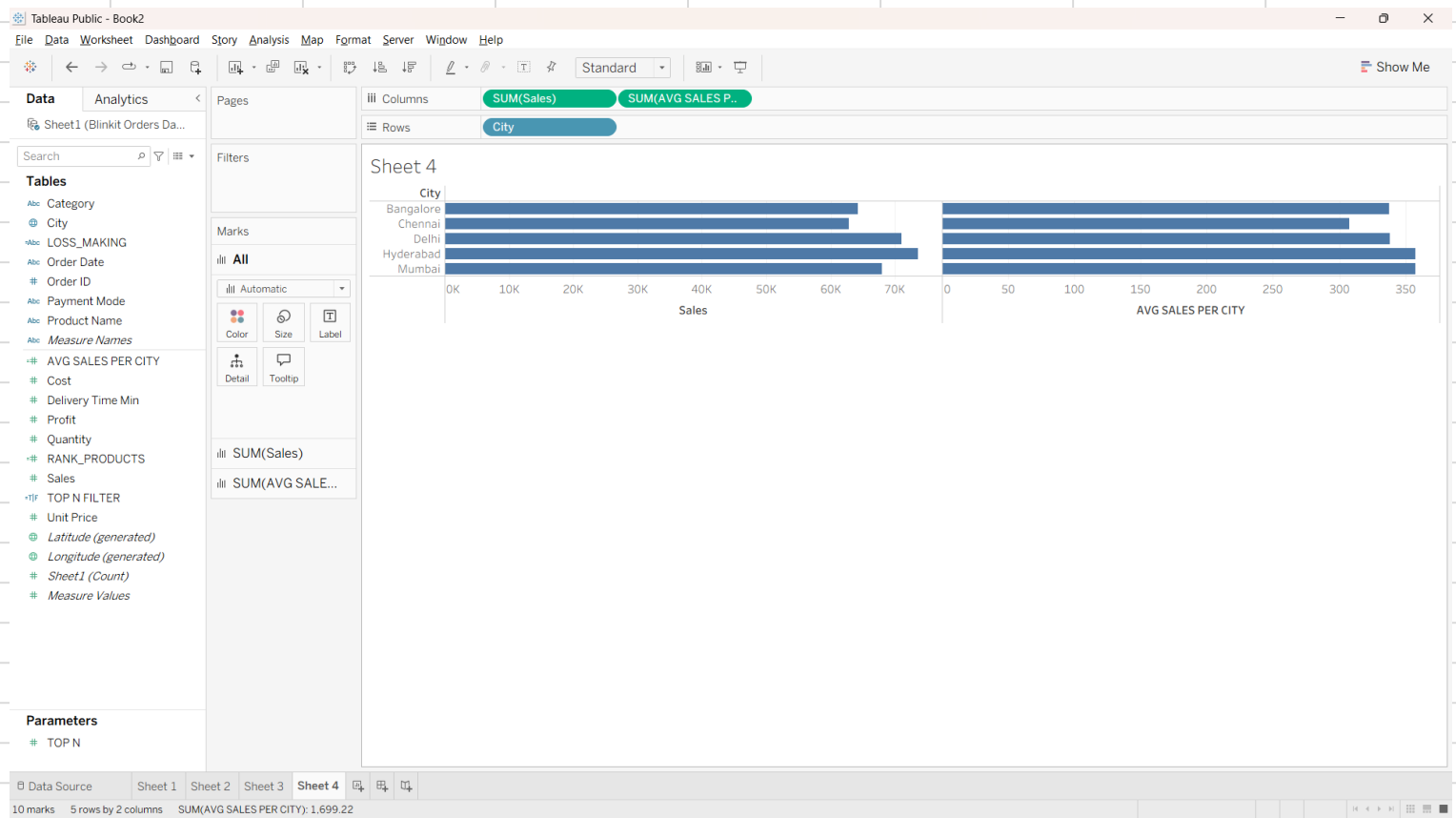
Task

Create a Top N Products dynamic analysis

ANSWER



QUESTION 8	Task 4: City-Level Benchmarking (FIXED LOD)
	Business Need
	Compare city performance independent of category filters
	Task
ANSWER	Calculate average sales per city using FIXED LOD.



QUESTION 9

Task 5: Customer/Product Detail Impact (INCLUDE LOD)

Business Need

The sales team wants average sales considering product-level detail, even when product is not shown.

Task

Create a calculation that finds the average sales at the product level, but allows the result to be used at a higher level (such as Category or Region). Use INCLUDE LOD.

ANSWER

